COST OF CAPITAL • COST OF EQUITY

The Dividend Growth Model

Estimating the Cost of Equity Using Real Indian Companies

$$Ke = D1/P0 + g$$

The core valuation equation behind this module

The Dividend Growth Model

Also known as the **Gordon Growth Model.** It values a share as the present value of all future dividends, assumed to grow at a constant rate forever.

FORMULA

$$P0 = D1 / (Ke - g)$$

REARRANGED FOR COST OF EQUITY

$$Ke = D1 / P0 + g$$

Model breaks down when g ≥ Ke — the denominator turns zero or negative.

VARIABLES

- **P0** — Current market price of the share
- **D1** — Expected dividend next year = $D_0 \times (1+g)$
- **D0** — Recent (current) dividend per share
- **Ke** — Cost of equity (required return)
- **g** — Constant expected growth rate of dividends

Example 1 — Hindustan Unilever (HUL)

Applying the CAPM-style DGM formula step by step

₹53.00

D_0 (FY2025 Dividend)

₹2,209

P_0 (Current Price)

14.35%

g (Growth Rate)

STEP 1 — Estimate D_1

$D_1 = 53 \times (1 + 0.1435) = ₹60.6$

STEP 2 — Apply the Formula

$Ke = 60.6/2209 + 0.1435 = 2.74\% + 14.35\%$

COST OF EQUITY (Ke)

= 17.1%

Required return demanded by HUL shareholders

HUL's dividend per share grew from ₹31.00 (FY2021) to ₹53.00 (FY2025).

Example 2 — Infosys Ltd.

Applying the same CAPM-style DGM formula

₹45.00

D_0 (FY2025 Dividend)

₹1,500

P_0 (Current Price)

10.7%

g (Growth Rate)

STEP 1 — Estimate D_1

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D1 = 45 × (1 + 0.107) = ₹49.8
```

STEP 2 — Apply the Formula

```
Ke = 49.8/1500 + 0.107 = 3.32% + 10.7%
```

COST OF EQUITY (Ke)

= 14.0%

Required return demanded by Infosys shareholders

Infosys' dividend grew from ~₹30 (FY2021) to ~₹45 (FY2025), including a special dividend in FY2024.

Growth Rate (g) Calculation

Two standard approaches used by analysts

1 Geometric Mean of Historical Dividends

$$g = (D_n / D_0)^{(1/n)} - 1$$

- D_0 = earliest dividend on record
- D_n = most recent dividend
- n = number of years between them

Reflects the company's actual dividend track record — but can be distorted by one-off special dividends.

2 Retention Ratio × Return on Equity

$$g = b \times r$$

- b (Retention Ratio) = 1 − Payout Ratio = 1 − DPS/EPS
- r (ROE) = return earned on reinvested equity

*Also called the **Sustainable Growth Rate** — reflects what the company can realistically repeat, based on reinvestment and profitability.*

Method 1: Geometric Mean Growth — Calculations

`g = (Dn / D0)^(1/n) − 1  |  4-year period (FY2021 → FY2025)`

HINDUSTAN UNILEVER

D0 (FY2021) = ₹31.00

Dn (FY2025) = ₹53.00

```
g = (53/31)^(1/4) − 1
g = (1.71)^0.25 − 1
```

g = 14.35%

INFOSYS

D0 (FY2021) ≈ ₹30.00

Dn (FY2025) ≈ ₹45.00

```
g = (45/30)^(1/4) − 1
g = (1.5)^0.25 − 1
```

g = 10.67%

Retention Ratio & Rate of Return (ROE)

The building blocks of the Sustainable Growth Rate model

RETENTION RATIO (b)

$$b = 1 - \text{Payout Ratio}$$

$$= 1 - (\text{DPS} / \text{EPS})$$

The proportion of earnings a company reinvests (keeps) rather than distributes as dividends.

Low b → mature company paying out most profits (e.g. FMCG)
High b → growth company reinvesting heavily

RATE OF RETURN (r / ROE)

$$r = \text{Net Income} / \text{Equity}$$

(Return on Equity)

The rate of return the company earns on the equity capital it reinvests back into the business.

Higher, more stable ROE → reinvested earnings compound faster → supports higher sustainable growth (g)

Retention Ratio & ROE — HUL Example

A mature FMCG company with a very high payout ratio

92.55%

Payout Ratio

7.45%

Retention Ratio (b)

21.55%

ROE (FY2025)

$$g = b \times r$$
$$g = 0.0745 \times 0.2155 = 1.6\%$$

SUSTAINABLE g

= 1.6%

KEY TEACHING INSIGHT

This sustainable estimate (1.6%) is far lower than the historical geometric-mean growth (14.35%). HUL's dividend history includes unusually large payouts in some years, so historical growth overstates what the company can sustainably repeat.

Retention Ratio & ROE — Infosys Example

A high-ROE IT services company with balanced capital allocation

68.5%

Payout Ratio

31.5%

Retention Ratio (b)

30.8%

ROE (3-yr avg)

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g = b × r
g = 0.315 × 0.308 = 9.7%
```

SUSTAINABLE g

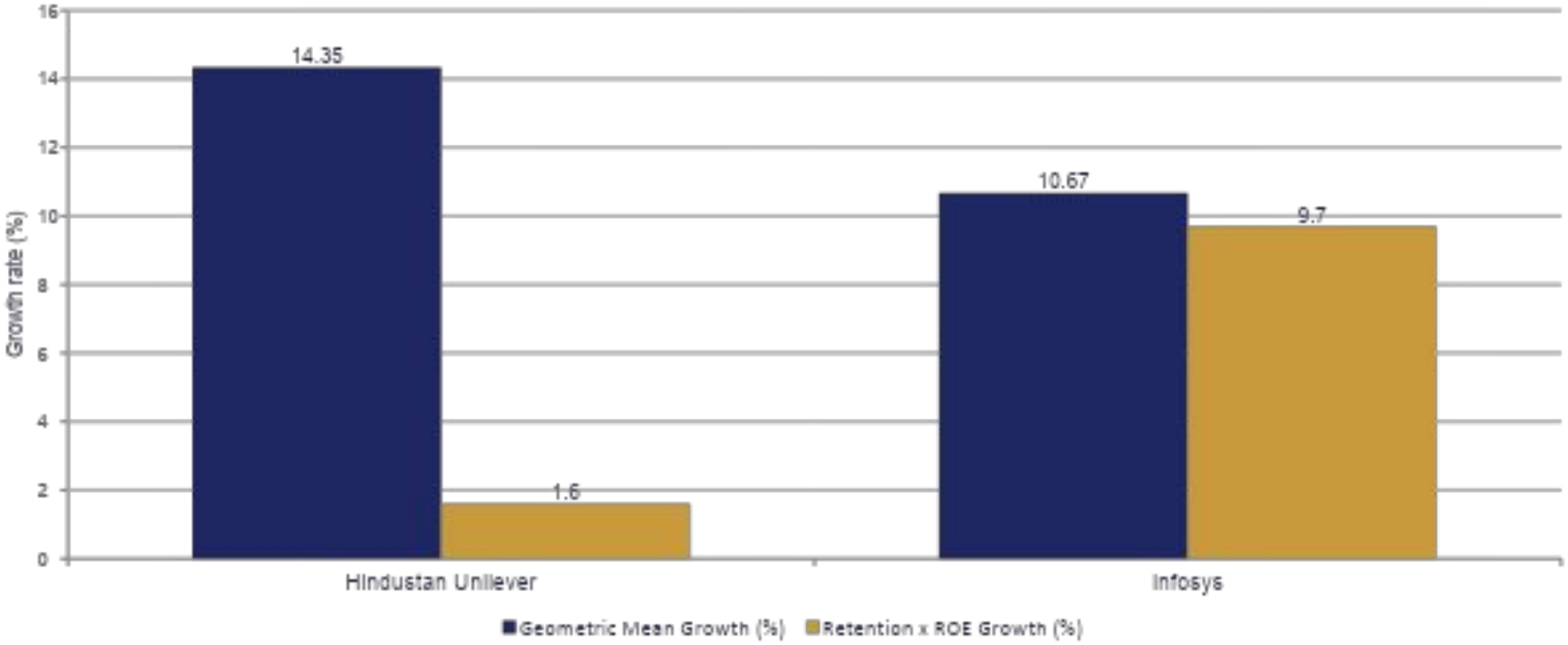
= 9.7%

KEY TEACHING INSIGHT

This sustainable estimate (9.7%) is close to the historical geometric-mean growth (10.67%) — Infosys' dividend policy has been relatively stable and consistent with its profitability, making the DGM estimate more reliable here.

Comparing the Two Growth Estimation Methods

Geometric Mean growth vs. Retention x ROE (Sustainable Growth) — HUL vs Infosys



Key Takeaways for Discussion

1

Stable payout → reliable DGM

When a company's payout ratio is stable (Infosys), both growth methods converge and the DGM estimate is more trustworthy.

2

Volatile payout → use judgment

When payout is volatile or unusually high (HUL), the two methods diverge sharply — analysts must judge which g to use, or apply a multi-stage growth model.

3

Watch the g ≥ Ke trap

The model breaks down entirely if g ≥ Ke: the denominator becomes zero or negative — always sanity-check before applying it mechanically.